

KEYSTONE

Dislocation-reversion in individual municipal bonds — buying forced-seller mispricings on the MSRB regulatory trade tape.

Asset class · US tax-exempt munis (CUSIP-level)

Backtest · 2013–2026

Benchmark · MUB (total return)

Universe · 1,416 bonds · 2.0M prints

+3.1%

EXCESS VS CTRL
p<0.001

73%

WIN RATE
1,350 trades

+4.4%

MEAN / TRADE
~1yr, incl coupon

+4.62%

CAGR
MUB +2.09%

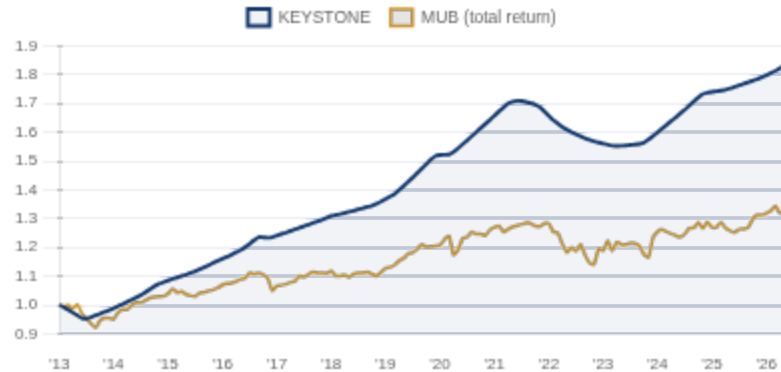
+82.9%

TOTAL (13.4Y)
MUB +31.8%

-9.2%

MAX DD*
MUB -13.7%

GROWTH OF \$1 — TOTAL RETURN VS MUB



	Total	CAGR	Max DD*
KEYSTONE strategy	+82.9%	+4.62%	-9.2%
MUB (total return)	+31.8%	+2.09%	-13.7%

REGIME ROBUSTNESS — EXCESS VS MATCHED CONTROL

Era	Trades	Win	Mean/tr	Excess	p
2013–2016	108	94%	+6.45%	+2.79%	<.001
2017–2019	63	94%	+7.40%	+2.17%	0.01
2020–2021	238	86%	+8.64%	+10.65%	<.001
2022	401	32%	-3.37%	-1.99%	1.00
2023–2025	564	85%	+6.07%	+4.22%	<.001

THE STRATEGY

Buy when a bond prints ≥ 3 pts below its own trailing 60-day median (a forced-seller dislocation) and it clears a trailing-liquidity gate. **Fill** at the next customer-buy print. **Hold ~1 year**; **exit** into a customer-sell print. Coupon accrues throughout.

The dealer bid–ask spread (~0.44 pt round-trip) kills short-term timing but is a one-time cost against growing carry + reversion. Buying idiosyncratic dislocations and holding to reversion beats random entry **in the same bonds** — pure timing alpha, net of spread, carry and selection.

SIGNAL STRENGTH BY DEPTH (FULL SAMPLE)

Entry threshold	Trades	Win	Excess	p
≥ 1.0 pt below median	2,267	70%	+0.44%	0.004
≥ 2.0 pt below median	1,786	72%	+1.63%	<.001
≥ 3.0 pt below median	1,350	73%	+3.10%	<.001

IN-SAMPLE → OUT-OF-SAMPLE (LOCKED)

Window	Trades	Mean/tr	Excess
In-sample 2012–2022	791	+2.55%	+2.07%
Out-of-sample 2023–25 (locked)	608	+6.32%	+4.45%

Known failure mode — sustained rate selloffs

Mean-reversion is short a rate shock. In 2022 the whole market repriced down and dislocations did not revert (~1.99% excess). Size accordingly; an optional MUB-trend overlay mitigates it.

DATA & VALIDATION

- Source: MSRB EMMA — the regulatory trade tape (every muni trade, tick-level price/yield/size/side).
- Cross-endpoint price/count match: **100%**; Δ price/ Δ yield corr ~0.98.
- Customer-buy $\geq$ customer-sell same-day: 96.9%.

CAPACITY & POSITIONING

- Deep dislocations are rare: ~100–250/yr across ~1,400 liquid bonds.
- Patient-capital, capacity-limited sleeve; complements a core muni book.
- Currently **dormant** — no bond ≥ 3 pt dislocated (calm market).

IMPLEMENTATION

- Daily screen on the fresh EMMA tape flags eligible dislocations.
- Fills sourced as the patient buyer into forced selling.
- Overlays: position sizing, credit screen, MUB-trend regime gate.